PROJECT: UNCOVERING THE WORLD'S OLDEST BUSINESSES

Staffelter Hof Winery is Germany's oldest business, established in 862 under the Carolingian dynasty. It has continued to serve customers through dramatic changes in Europe, such as the Holy Roman Empire, the Ottoman Empire, and both world wars. What characteristics enable a business to stand the test of time?

To help answer this question, BusinessFinancing.co.uk researched the oldest company still in business in **almost** every country and compiled the results into several CSV files. This dataset has been cleaned.

Having useful information in different files is a common problem. While it's better to keep different types of data separate for data storage, you'll want all the data in one place for analysis. You'll use joining and data manipulation to work with this data and better understand the world's oldest businesses.

The Data

`data/businesses.csv` and `data/new_businesses.csv`

Column	Description
`business`	Name of the business (varchar)
`year_founded`	Year the business was founded (int)
`category_code`	Code for the business category (varchar)
`country_code`	ISO 3166-1 three-letter country code (char)

`data/countries.csv`

Column	Description
`country_code`	ISO 3166-1 three-letter country code (varchar)
`country`	Name of the country (varchar)
`continent`	Name of the continent the country exists in (varchar)

`data/categories.csv`

Column	Description
`category_code`	Code for the business category (varchar)
`category`	Description of the business category (varchar)

```python
# Import necessary libraries
import pandas as pd

# Load the data
businesses = pd.read_csv("data/businesses.csv")
new_businesses = pd.read_csv("data/new_businesses.csv")
countries = pd.read_csv("data/countries.csv")
categories = pd.read_csv("data/categories.csv")
```

```
# Start coding here!
# Use as many cells as you like
# What is the oldest business on every continent?
# Start by merging the businesses and countries datasets into one
businesses_countries = businesses.merge(countries, on="country_code")

# Create a new DataFrame that lists only the continent and oldest year_founded
continent = businesses_countries.groupby("continent").agg({"year_founded":"min"})

# Merge this continent DataFrame with businesses_countries
merged_continent = continent.merge(businesses_countries, on=["continent",
"year_founded"])

# Subset the continent DataFrame so that only the four columns of interest are
included, saving it as oldest_business_continent
oldest_business_continent = merged_continent[["continent", "country", "business",
"year_founded"]]

# View the result
print(oldest_business_continent)

# How many countries per continent lack data on the oldest businesses?
# Does including the `new_businesses` data change this?

# Add the data in new_businesses to the existing businesses
all_businesses = pd.concat([new_businesses, businesses])

# Perform a new merge between the businesses and the countries data. Use additional
parameters this time to perform an outer merge and create an indicator column to
better see the missing values. An outer merge combines two DataFrames based on a key
column and includes all rows from both DataFrames
new_all_countries = all_businesses.merge(countries, on="country_code", how="outer",
indicator=True)

# Filter to find countries with missing business data
new_missing_countries = new_all_countries[new_all_countries["_merge"] != "both"]

# Group by continent and create a "count_missing" column
count_missing = new_missing_countries.groupby("continent").agg({"country":"count"})
count_missing.columns = ["count_missing"]

# View the results
print(count_missing)

# Which business categories are best suited to last over the course of centuries?

# Start by merging the businesses and categories data into one DataFrame
```

```
businesses_categories = businesses.merge(categories, on="category_code")

# Merge all businesses, countries, and categories together
businesses_categories_countries = businesses_categories.merge(countries,
on="country_code")

# Create the oldest by continent and category DataFrame
oldest_by_continent_category = businesses_categories_countries.groupby(["continent",
"category"]).agg({"year_founded":"min"})
oldest_by_continent_category.head()
```

```
       continent    country                      business  year_founded
0         Africa  Mauritius                Mauritius Post          1772
1           Asia      Japan                    Kongō Gumi           578
2         Europe    Austria  St. Peter Stifts Kulinarium           803
3  North America     Mexico  La Casa de Moneda de México          1534
4        Oceania  Australia                Australia Post          1809
5  South America       Peru      Casa Nacional de Moneda          1565
               count_missing
continent
Africa                     3
Asia                       7
Europe                     2
North America              5
Oceania                   10
South America              3
```

continent	category
Africa	Agriculture
Africa	Aviation & Transport
Africa	Banking & Finance
Africa	Distillers, Vintners, & Breweries
Africa	Energy

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